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Editor

dspexp1amlx.mlx | exp1g.m | exp1bg.m | exp1cg.m | exp2g.m | exp3g.m | exp5.m

/MATLAB Drive/exp5.m

```
clc;
clear;
close all;

BWcc = 20;              % MHz per component carrier
SE = 4;                 % bits/s/Hz
CC = [1 2 3];           % Number of component carriers

AggBW = BWcc .* CC;     % MHz
Throughput = AggBW .* SE; % Mbps

disp('CC Aggregated BW (MHz) Estimated Throughput (Mbps)')
disp([CC' AggBW' Throughput'])

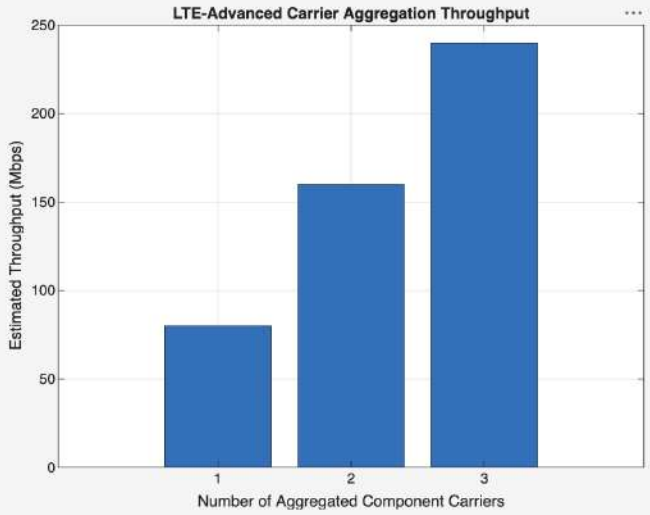
figure;
bar(CC, Throughput);
grid on;

xlabel('Number of Aggregated Component Carriers');
ylabel('Estimated Throughput (Mbps)');
title('LTE-Advanced Carrier Aggregation Throughput');

% Expected output:
% The graph increases with the number of aggregated component carriers.
% 1 carrier provides 80 Mbps.
% 2 carriers provide 160 Mbps.
% 3 carriers provide 240 Mbps.
```

Figures

Figure 1



Command Window

```
CC Aggregated BW (MHz) Estimated Throughput (Mbps)
     1    20    80
     2    40   160
     3    60   240

>>
```

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Editor

dspexp1amlx.mlx | exp1g.m | exp1bg.m | exp1cg.m | exp2g.m | exp3g.m | exp5.m | exp5b.m

/MATLAB Drive/exp5b.m

```
clc;
clear;
close all;

% Carrier bandwidths in MHz
BW = [5 10 15 20];

% Spectral efficiency in bits/s/Hz
SE = 4;

% Calculate estimated throughput in Mbps
TP = BW .* SE;

% Display results
disp('Bandwidth (MHz)   Throughput (Mbps)');
disp([BW' TP']);

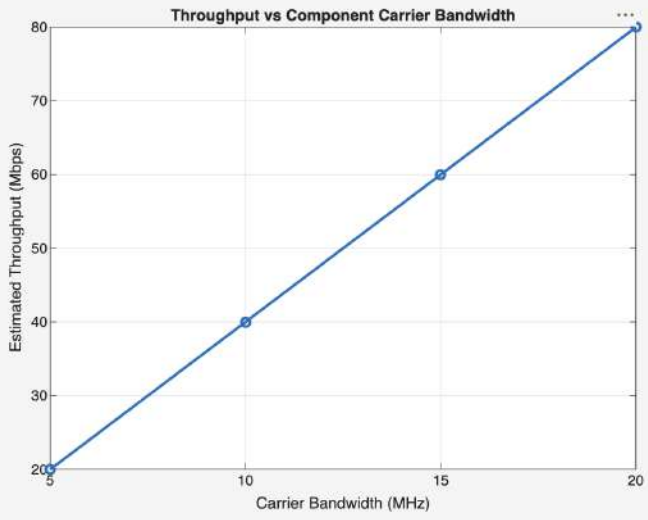
% Plot throughput
figure;
plot(BW, TP, 'o-', 'LineWidth', 2);
grid on;

xlabel('Carrier Bandwidth (MHz)');
ylabel('Estimated Throughput (Mbps)');
title('Throughput vs Component Carrier Bandwidth');

% Expected Output:
% Throughput increases as component-carrier bandwidth increases.
```

Figures

Figure 1



Command Window

```
Bandwidth (MHz)   Throughput (Mbps)
     5    20
    10    40
    15    60
    20    80

>>
```